

# Danshu Jiang

Biotechnology Undergraduate | Aspiring Bioinformatician

Changsha, Hunan (CN)

✉ [danshujiang0@gmail.com](mailto:danshujiang0@gmail.com)

🌐 [danshujiang](#)

## SUMMARY

Undergraduate Biotechnology student at Hunan Agricultural University with hands-on research experience in plant tissue culture, virus detection, and virus-free technology design. Passionate about bridging biology and computation—applying Python and AI tools to solve biological problems. Seeking opportunities in bioinformatics to combine wet-lab expertise with computational analysis.

## SKILLS

### Programming & Bioinformatics

Python   Biopython   Pandas   Data Analysis  
Sequence Analysis

### AI & Productivity Tools

ChatGPT   Claude Code   Large Language Models  
AI-assisted Research   Prompt Engineering

### Wet-Lab Techniques

RT-PCR   RNA/DNA Extraction   Gel Electrophoresis  
Tissue Culture   Aseptic Technique   Virus Detection

### Collaboration & Version Control

GitHub   Git   Scientific Writing   Protocol Documentation

## PROJECTS (5)

### Virus-Free Technology for Longya Lily (*Lilium brownii*)

Plant Tissue Culture   Thermotherapy   Chemotherapy   Meristem Culture

Designed and optimized a virus-free propagation protocol for Longya Lily, an economically important cultivar. Integrated meristem tip culture with thermotherapy and chemotherapy to eliminate viral pathogens.

- Established shoot tip isolation and meristem culture protocols for *Lilium brownii*
- Applied thermotherapy (37°C, 4 weeks) combined with ribavirin chemotherapy to enhance virus elimination efficiency
- Achieved regeneration of virus-free plantlets with improved vigor compared to infected donor material
- Designed the experimental workflow from explant sterilization through acclimatization

### Virus Detection in Lily — LMoV, LSV & CMV

RT-PCR   RNA Extraction   Gel Electrophoresis   Viral Diagnostics

Screened lily samples for three major viral pathogens (Lily Mottle Virus, Lily Symptomless Virus, Cucumber Mosaic Virus) using molecular diagnostic techniques. Validated virus-free status of regenerated plants.

- Performed RT-PCR detection of LMoV, LSV, and CMV from leaf tissue samples

- Extracted total RNA and synthesized cDNA for multiplex screening
- Compared virus incidence rates between field-collected and in-vitro-regenerated material
- Documented detection protocols and maintained experimental records for reproducibility

## Plant Tissue Culture & Regeneration Systems

Aseptic Technique   Media Preparation   Micropropagation   Greenhouse Acclimatization

Established and maintained in-vitro culture systems for lily and other horticultural crops. Investigated the effects of plant growth regulators on shoot proliferation and root induction.

- Prepared MS media with varying concentrations of 6-BA, NAA, and IBA for organogenesis
- Monitored and controlled aseptic conditions to minimize contamination rates below 5%
- Optimized subculture intervals and rooting protocols across multiple lily genotypes
- Acclimatized tissue-cultured plantlets in greenhouse conditions with >85% survival rate

## TNF Gene Bioinformatics Analysis

Python   Biopython   NCBI   Sequence Alignment   Phylogenetic Analysis   Protein Structure Prediction

Performed computational analysis of Tumor Necrosis Factor (TNF) gene across multiple species. Applied bioinformatics tools for sequence alignment, conserved domain identification, and phylogenetic analysis to understand evolutionary relationships and functional implications.

- Retrieved TNF gene sequences from NCBI GenBank across multiple species
- Performed multiple sequence alignment using Biopython and Clustal Omega
- Identified conserved domains and predicted protein functional sites
- Constructed phylogenetic tree to analyze TNF gene evolutionary relationships across species

## AI-Assisted Experimental Design for Lily Virus Elimination

ChatGPT   Claude Code   GitHub   LLM   Primer Design   Data Analysis   Scientific Writing

Leveraged large language models (ChatGPT, Claude) to optimize virus elimination protocols for lily. Applied AI tools for literature review, primer design, data analysis, and experimental documentation, significantly accelerating the research workflow.

- Used ChatGPT and Claude Code to screen and summarize relevant virology literature, reducing literature review time by 60%
- Applied LLM-assisted analysis of LMoV/LSV/CMV genome sequences to guide RT-PCR primer design
- Maintained version-controlled experimental protocols and data records on GitHub
- Employed AI tools for statistical analysis, figure generation, and scientific writing

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### EDUCATION (1)

Sept 2022 - Jun 2026

**Bachelor of Science Biotechnology** at [Hunan Agricultural University](#)

Molecular Biology   Genetics   Plant Tissue Culture   Virology   Bioinformatics   Biochemistry   Cell Biology   Microbiology

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### LANGUAGES (2)

Chinese

Native speaker



English

Professional working proficiency



INTERESTS (2)

Bioinformatics

- Genomic Data Analysis
- Transcriptomics
- Viral Genomics
- Computational Biology

Plant Biotechnology

- Virus-Free Propagation
- Molecular Breeding
- Germplasm Conservation